INCH-POUND

MIL-C-70950 (AR)
12 January 1990

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M148A1B1E1

This Specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements, quality assurance provisions, and packaging for the 105mm, M148A1B1E1 Spiral-Wrapped Cartridge Case (SWCC).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-46487 - Cleaning, Preparation and Organic Coating of Steel Cartridge Cases
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

7548235 - Cartons, Ammunition, for Interplant Shipment of Steel Cartridge Cases, Empty 37mm to 120mm, Inclusive
8647040 - Gage, Chamber
11735736 - Case, Cartridge, 105mm, M148A1B1E1

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ASTM Methods - AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B117 - Standard Method of Salt Spray (Fog) Testing
ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The cartridge case shall comply with the requirements of the applicable drawings and referenced specifications.

3.2 Material. The supplier shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material used in manufacture.

3.3 Mechanical properties. The mechanical properties specified on drawings and referenced specifications shall be uniform throughout each section.

3.4 Protective coating. Protective coating as specified on drawings and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration and shall be uniform and continuous. Scratches exposing base metal are not permitted.

3.5 Assembly tightness. The case shall be assembled in such a manner that there is no relative rotation between the base, the cup, and the body when a torque of 75±5 in.-lbs. is applied to the base.

3.6 Maximum profile and alignment. Maximum profile and alignment of the cartridge case shall comply with chamber gage drawing 11735736.

3.7 Ballistics. The cartridge case shall comply with the requirements of 3.7.1 through 3.7.5 when subjected to ballistic testing in accordance with 4.6.7 in the following phases:

a. Excess pressure phase
b. Cold temperature phase
c. Excess, excess pressure phase (for information only).

3.7.1 Obturation. The cartridge case shall effectively seal the gun breech against leakage of gases generated during firing.

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3.7.2 Extraction. After firing, the cartridge case shall be extracted by the service extractors when the gun is operated in the normal manner.

3.7.3 Primer retention. The cartridge case shall retain its primer during firing and extraction.

3.7.4 Propellant gas leakage. The cartridge case shall effectively prevent the entry of propellant gas through or between case components into the gun chamber.

3.7.5 Cracks or splits. After firing, the cartridge case shall be free of cracks or splits through all body plies.

3.8 Workmanship. The requirements for workmanship are as specified on drawings, referenced specifications and the following:

3.8.1 Metal defects. The cartridge case shall be free of cracks, blisters, laps, deep scratches, dents, embedded foreign matter, porosity, and other metal defects.

3.8.2 Burrs. No cartridge case shall have any burrs.

3.8.3 Foreign matter. No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to accept defective material.

4.3 Classification of inspections. The following types of inspections shall be performed on this item:

a. First article inspection

b. Quality Conformance Inspection

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of this specification. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency, shall also be submitted whenever a change occurs in specification, or source of supply, such as to significantly affect product uniformity as determined by the responsible government technical agency. The first article sample shall consist of the following items in quantities indicated.

Part Description	Drawings	Quantity
Cup (uncoated)	11735732	4
Cup (coated)	11735732	10
Collar (uncoated)	11735733	4
Collar (coated)	11735733	10
Base (uncoated)	11735735	4
Base (coated)	11735735	10
Test Panels (See MIL-C-46487)	--------	2
Cartridge Case Body Blank	--------	4
Cartridge Case (M148A1B1E1)	11735736	254

4.4.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.4.3 Rejection. See MIL-A-48078

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Table I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70950 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	M148A1B1E1 Cartridge Case Assembly	1 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cup (uncoated)			
	Dwg. 11735732			
	Hardness	4 units	3.3	
	Collar (uncoated)			
	Dwg. 11735733			
	Hardness	4 units	3.3	
	Base (uncoated)			
	Dwg. 11735735			
	Hardness	4 units	3.3	
	Cup (coated)			
	Dwg. 11735732			
	Examination for defects	10 units	3.1	
	Coating thickness	4 units	3.4	
	Salt spray	2 units	3.4	
	Collar (coated)			
	Dwg. 11735733			
	Examination for defects	10 units	3.1	
	Coating thickness	4 units	3.4	
	Salt spray	2 units	3.4	

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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Table I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70950 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2	DRAWING NUMBER
	M148A1B1E1 Cartridge Case Assembly			See Below
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base (coated)			
	Dwg. 11735735			
	Examination for defects	10 units	3.1	
	Coating thickness	4 units	3.4	
	Salt spray	2 units	3.4	
	Test panels (see MIL-C-46487)			
	Weight of phosphate coating	2 units	3.4	
	Flexibility	2 units	3.4	
	Cartridge case body blank			
	Hardness	4 units	3.3	
	Coating thickness	4 units	3.4	
	Salt spray	2 units	3.4	
	Cartridge case			
	Dwg. 11735736			
	Examination for defects	254 units	3.1	
	Distance from base O.D. to edge of body plies	2 units	3.1	
	Salt spray	2 units	3.4	
	Ballistics test		3.7	
	Excess pressure	100 units		
	Cold temperature	50 units		
	Excess, excess pressure (info only)	100 units		

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.5 Quality conformance inspection.

4.5.1 Lot formation. Lots shall comply with the lot formation provisions of MIL-A-48078. In addition, one lot of cartridge cases shall consist of not more than 25,000 units.

4.5.2 Examinations and tests.

4.5.2.1 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Where specified herein, attributes sampling inspection shall be conducted in accordance with Table II below. Samples shall be selected at random.

Table II. Sampling

Lot Size	Inspection Level	
	I Major Defects	II Minor Defects
2 to 13	*	*
14 to 90	*	13
91 to 150	125	13
151 to 500	125	30
501 to 1200	125	35
1201 to 3200	125	43
3201 to 10000	125	50
10001 to 35000	296	61

Numbers under inspection levels indicate sample size.

* Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for both inspection levels.

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4.5.2.2 Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.5.3 Tool control dimensions. Dimensions marked "Tool Control" shall be gaged at the beginning or end of each (24 hour) working day, on each machine in operation and for the first component produced after any tooling change, adjustment or repair. When destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.5.4 Inspection equipment. Inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the tables starting with paragraph 4.5.5.

4.5.5 Classification of characteristics.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70950 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.1	Cup	1 OF 1	11735732
			NEXT HIGHER ASSEMBLY 9323901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Outside diameter	Table II	3.1	Gage
102	Inside diameter	Table II	3.1	Gage
103	Length, total	Table II	3.1	Gage
104	Diameter of recess	Table II	3.1	Gage
105	Angle of recess	Table II	3.1	Gage
106	Thickness of lip	Table II	3.1	Gage
107	Taper on diameter	Table II	3.1	Gage
108	Hardness	4.5.6.1	3.3	4.6.1.3
109	Thickness	Table II	3.1	Gage
110	Protective coating missing or inadequate	Table II	3.4	Visual
111	Surface finish	Table II	3.1	Visual
112	True position of I.D.	Table II	3.1	Gage
113	Radius at corner	Table II	3.1	Gage
114	Coating thickness	4.5.6.2.1	3.4	Gage
115	Salt spray	4.5.6.2.2	3.4	4.6.2.1
Minor				
201	Poor workmanship	Table II	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.2	Collar	1 OF 1	11735733
			NEXT HIGHER ASSEMBLY: 9323901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Boss outside diameter	Table II	3.1	Gage
102	Width of boss	Table II	3.1	Gage
103	Wall thickness of boss (two places)	Table II	3.1	Gage
104	Depth of boss groove	Table II	3.1	Gage
105	Large outside diameter	Table II	3.1	Gage
106	Circumferential taper angle of boss	Table II	3.1	Gage
107	Circumferential taper depth of boss	Table II	3.1	Gage
109	Radius	Table II	3.1	Gage
110	True position of outside diameter	Table II	3.1	Gage
111	Large radius of flange	Table II	3.1	Gage
112	Radius between boss and flange	Table II	3.1	Gage
113	Hardness	4.5.6.1	3.3	4.6.1.2
114	Protective coating missing or inadequate	Table II	3.4	Visual
115	Surface finish	Table II	3.1	Visual
116	Coating thickness	4.5.6.2.1	3.4	Gage
117	Salt spray	4.5.6.2.2	3.4	4.6.2.1
Minor				
201	Small radius of flange	Table II	3.1	Gage
202	Poor workmanship	Table II	3.8	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.3	Base	1 OF 2	11735735
			NEXT HIGHER ASSEMBLY 9323901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Large outside diameter	Table II	3.1	Gage
102	Small outside diameter	Table II	3.1	Gage
103	True position of small outside diameter	Table II	3.1	Gage
104	Diameter of outside edge of channel	Table II	3.1	Gage
105	True position of outside edge of channel diameter	Table II	3.1	Gage
106	Width of channel	Table II	3.1	Gage
107	Angle of channel walls	Table II	3.1	Gage
108	Pitch diameter of thread, max.	Table II	3.1	Gage
109	Minor diameter of thread, max.	Table II	3.1	Gage
110	True position of pitch diameter	Table II	3.1	Gage
111	Diameter of thread relief	Table II	3.1	Gage
112	True position of thread relief diameter	Table II	3.1	Gage
113	Diameter of counterbore	Table II	3.1	Gage
114	True position of counterbore diameter	Table II	3.1	Gage
115	Depth of thread relief	Table II	3.1	Gage
116	Depth of counterbore	Table II	3.1	Gage
117	Distance from flange to bottom of counterbore	Table II	3.1	Gage
118	Height of channel protrusion	Table II	3.1	Gage
119	Width of channel protrusion	Table II	3.1	Gage

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.3	Base	2 OF 2	11735735
			NEXT HIGHER ASSEMBLY 9323901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	(cont'd)			
120	Overall thickness of base	Table II	3.1	Gage
121	Depth of channel	Table II	3.1	Gage
122	Thickness of flange	Table II	3.1	Gage
123	Hardness	4.5.6.1	3.3	4.6.1.1
124	Surface finish	Table II	3.1	Visual
125	Protective coating missing or inadequate	Table II	3.4	Visual
126	Coating thickness	4.5.6.2.1	3.4	Gage
127	Salt spray	4.5.6.2.2	3.4	4.6.2.1
128	Metal defects	100%	3.8.1	4.6.3
Minor				
201	Max. offset of the diameter of the outside surface of the channel at 45°	Table II	3.1	Gage
202	Max. offset of the diameter of the inside surface of the groove at 45°	Table II	3.1	Gage
203	Radii or chamfers missing	Table II	3.1	Visual
204	Poor workmanship	Table II	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.4	M148A1B1E1 Cartridge Case Assembly	1 OF 2	11735736
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Thickness of body material prior to wrapping	Table II	3.1	Gage
102	Total length	Table II	3.1	Gage
103	Distance from flange (base) to first basic diameter	Table II	3.1	Gage
104	Distance from flange (base) to second basic diameter	Table II	3.1	Gage
105	Distance from flange (base) to third basic diameter	Table II	3.1	Gage
106	Distance from flange (base) to fourth basic diameter	Table II	3.1	Gage
107	Inside diameter of mouth	Table II	3.1	Gage
108	True position of mouth I.D.	Table II	3.1	Gage
109	O. D. of mouth	Table II	3.1	Gage
110	True position of outside diameter of mouth	Table II	3.1	Gage
111	Improper number of wraps at mouth	Table II	3.1	Visual
112	Protective coating missing or inadequate	Table II	3.4	Visual
113	Minimum radius of sheet edge	Table II	3.1	Visual
114	Location of oval indents from mouth	Table II	3.1	Gage
115	Angular location of oval indents to one another	Table II	3.1	Gage

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.5.4	M148A1B1E1 Cartridge Case Assembly	2 OF 2	11735736
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	(cont'd)			
116	Location of weld to edge of inner wrap	Table II	3.1	Gage
117	Length of each oval indent	Table II	3.1	Gage
118	Width of each oval indent	Table II	3.1	Gage
119	Hardness of body blank	4.5.6.1	3.3	4.6.1.4
120	Assembly tightness	100%	3.5	4.6.5
121	Metal defects	100%	3.8.1	4.6.3
122	Salt spray (body blank)	4.5.6.2.2	3.4	4.6.2.1
123	Salt spray (assembly)	4.5.6.2.2	3.4	4.6.2.1
124	Distance from base O.D. to outer ply	4.5.6.3	3.1	4.6.6
125	Distance from base O.D. to two inner plies	4.5.6.3	3.1	4.6.6
126	Location of weld to edge of outer wrap	Table II	3.1	Gage
127	Profile and alignment	100%	3.6	Gage
128	Ballistics	4.5.6.4.2	3.7	4.6.7.2
Minor				
201	Radii missing	Table II	3.1	Visual
202	Stamping incorrect or illegible	Table II	3.1	Visual
203	Location of spot welds from mouth	Table II	3.1	Visual
204	45° overlap of body wraps at mouth	Table II	3.1	Gage
205	Poor workmanship	Table II	3.8	Visual

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4.5.6 Testing. This section describes sampling plans, criteria for acceptance or rejection of individual samples, and criteria for acceptance or rejection of the entire lot.

4.5.6.1 Hardness. Ten samples shall be randomly selected for testing. Failure of one hardness reading is cause for rejection of the sample. Failure of one sample is cause for rejection of the lot.

4.5.6.2 Protective coating.

4.5.6.2.1 Coating thickness. Samples shall be selected in accordance with Table II. Failure to comply with the thickness requirement in the component drawing shall be cause for rejection of the sample. Failure of one sample shall be cause for rejection of the lot.

4.5.6.2.2 Salt spray. Five samples shall be randomly selected for testing. Samples shall be evaluated at the end of the test period in accordance with the detail specification for the protective finish. The cartridge case assemblies subjected to the salt spray test shall show no more than 1/8 inch creepage, blistering, or loss of adhesion of the coating from the scribe mark. At all other points, there shall be no more than a trace of film failure (rust grade 9 ASTM D-610), and not more than 5 scattered blisters none larger than 1mm (3/64 inch) in diameter on a 48 square inch area. Rust within one thirty second (1/32) inch of all other edges shall be permitted. Failure to comply shall be cause for rejection of the sample. Rejection of one sample shall be cause for rejection of the lot.

4.5.6.3 Distance from base O.D. to edge of body plies. In order to determine the integrity of the assembly, two samples shall be inspected during each shift or eight hour production period. One sample shall be from the middle and one from the end of the shift. If any change is made to the assembly process, another sample shall be inspected. Failure of any sample to comply with requirements shall be cause for rejection of all production subsequent to the last acceptable inspection.

4.5.6.4 Ballistics. Physical evidence of noncompliance with the applicable table below, with or without video indications is cause for rejection. Video detection of emissions from the breach area alone is insufficient cause for rejection.

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4.5.6.4.1 First article. The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III.

Table III

Acceptance Criteria For First Article Ballistics Test

Defect Characteristic	Accept	Reject
1. Obturation failure	0	1
2. Extraction failure	0	1
3. Primer retention failure	0	1
4. Propellant gas leakage (Base burn mark)	0	1
5. Propellant gas leakage (Cup burn mark)	0	1
6. Propellant gas leakage (Body burn mark)	3	4
7. Cracks or splits through all Body plies	3	4

4.5.6.4.2 Production lot. One hundred cartridge cases from each lot will be ballistically tested in the excess pressure phase. Acceptance and rejection criteria shall be in accordance with Table IV. If test firing experiences an extraction or primer retention failure, the lot is eligible for retest firing of an additional sample of two hundred cases. The cumulative sample of three hundred must now meet the requirements of Table IV.

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Table IV

Acceptance Criteria For Production Lot Ballistics Test

Defect Characteristic	First Submission Accept	First Submission Reject	Cumulative Submission Accept	Cumulative Submission Reject
1. Obturation failure	0	1	Not permitted	
2. Extraction failure	0	2	1	2
3. Primer retention fail.	0	2	1	2
4. Propellant gas leakage (Base burn mark)	0	1	Not permitted	
5. Propellant gas leakage (Cup burn mark)	0	1	Not permitted	
6. Propellant gas leakage (Body burn mark)	2	3	Not permitted	
7. Cracks or splits through all Body plies	2	3	Not permitted	

4.6 Methods of inspection.

4.6.1 Hardness test. Hardness tests shall be conducted on components in accordance with ASTM E18 prior to cartridge case assembly.

4.6.1.1 Base. Seven hardness values shall be taken across the center of the rear surface of the base on both sides of the primer hole at approximately 1/4 inch intervals.

4.6.1.2 Collar. Four hardness values shall be taken on the underside of the flange of the collar approximately 90° apart.

4.6.1.3 Cup. Four hardness values shall be taken on the cylindrical surface of the cup, excluding the flange, approximately 90° apart.

4.6.1.4 Body blank. Hardness values shall be taken at approximately 12 inch intervals both vertically and horizontally through the center of the length and width of the body blank.

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4.6.2 Salt spray. Salt spray test shall be performed utilizing procedures specified in ASTM B117 and in accordance with the applicable detailed specification for the protective finish, except the salt exposure periods and solution shall be in accordance with the component drawings.

4.6.3 Metal defects detection. Four power (4X) magnification shall be used in the examination of the base and the exterior surface of the case assembly. The interior of the case assembly may be examined without magnification.

4.6.4 Maximum profile and alignment. Testing shall be conducted after application of protective coating. The case shall freely enter a maximum profile and alignment gage, Dwg. 8647040. The gate of the gage shall close without interference with the cartridge case base.

4.6.5 Assembly tightness. The cartridge case shall be held in an appropriate fixture and 75±5 in-lbs of torque shall be applied in both directions between the base and body. The moment arm shall be normal to the centerline of the case. Observation shall be made for any relative motion between the base, the cup and the body.

4.6.6 Distance from base O.D. to edge of body plies. Samples shall be cut along a center line of the case and gaged for minimum positioning depth of all body plies between the collar and cup. The sectioned case shall also be inspected for evidence of metal defects and poor workmanship.

4.6.7 Ballistics test. A color video camera shall be aimed at the breach of the gun and operated in such a manner as to detect any evidence of propellant gas leakage during firing. Video shall be used as an aid in detection of propellant gas leakage and inadequate obturation by identifying suspect cases. Video recordings will not be used as accept reject criteria. Camera exposure rate shall be not less than 30 frames per second. Provisions such as metal or canvas drapes shall be in place to shield the camera from blinding caused by muzzle flash. To prevent the recording and interpretation of extraneous video signals as propellant gas leakage during ballistic testing, the gun must be free of all lubricants and rust preventives in the breech and chamber areas. If such materials have been previously applied, the gun shall be wiped clean and warmer rounds fired to clear any residue. All cases shall be examined immediately after firing for compliance with defect characteristics listed in Tables III or IV as applicable.

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4.6.7.1 First article sample.

4.6.7.1.1 Test phases.

4.6.7.1.1.1 Excess pressure phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure with the following limits. Minimum mean value of the test rounds fired to be 66,000 psi (455.00 MPa) piezo equivalent pressure and a maximum individual pressure to be 68,500 psi (472.29 MPa) piezo equivalent pressure. The complete rounds will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5°C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.1.2 Cold temperature phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 57,900 psi (399.20 MPa) to 60,900 psi (419.90 MPa) piezo equivalent pressure, when temperature conditioned at 70°F. The complete rounds will be temperature conditioned at -35°F + 5°F (-37°C + 3°C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.1.3 Excess, excess pressure phase. The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to produce a chamber pressure of 72,300 psi (498.49 MPa) to 74,200 psi (511.49 MPa) piezo equivalent pressure. The cartridge will be will be temperature conditioned at +70°F ± 2.5°F (+21°C ± 1.5 °C) for 24 hours minimum and fired in a serviceable gun for which the round is standard. The gun will be operated in the service manner.

4.6.7.1.4 Failure modes.

4.6.7.1.4.1 Obturation. The case shall be examined for evidence of obturation failure, i.e. failure of the cup to expand sufficiently to prevent passage of propellant gas to the rear of the chamber enclosing the case base. This failure is evidenced by presence of burn mark(s) on the cup or base.

4.6.7.1.4.2 Extraction. Cartridge cases which failed to completely eject shall be examined for sufficient looseness in the gun chamber to permit removal by hand.

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4.6.7.1.4.3 Primer retention. The case shall be examined for retention of its primer after firing.

4.6.7.1.4.4 Propellant gas leakage. The cartridge case shall be examined for entry of propellant gas into the gun chamber by passing through any component or component interface. Mechanisms of failure include but are not limited to burn-through, rupture, joint failure, cracks and gas flow between body plies. Leakage is evidenced by the presence of burn marks on the component causing failure.

4.6.7.1.4.5 Cracks or splits. The cartridge case shall be visually examined for cracks or splits.

4.6.7.2 Production lot. Production lot acceptance tests shall be conducted in the Excess Pressure Phase. See 4.6.7.1.1.1 herein.

5. PACKAGING

5.1 Preservation and packaging. Packaging shall be as specified on Dwg. 7548235.

5.2 Packing and marking. Packing and marking shall be as specified on Dwg. 7548235 and MIL-STD-1169.

6. NOTES. (This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This cartridge case is intended to be used in the 105mm, target practice tank ammunition.

6.2 Process control lots. Prior to fabrication of a first article sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to five process control lots. These process control lots shall consist of not less than ten nor more than twenty-five cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of these lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.3 Visual examination qualification. When compliance with requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.4 Ordering data. See MIL-A-48078.

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6.5 Submission of firing records and ammunition data cards. In addition to the normal distribution of records, when the cartridge case is procured by the Army Armament, Munitions and Chemical Command, one copy of all firing records and ammunition data cards shall be forwarded to:

Commander
U. S. Army Armament Research, Development, and Engineering Center
ATTN: AMSMC-QAT-A (D)/SMCAR-CCH-P
Picatinny Arsenal, NJ 07806-5000

6.6 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to:

Commander
U. S. Army Armament Research, Development, and Engineering Center
ATTN: AMSMC-QAT-I (D)
Picatinny Arsenal, NJ 07806-5000

6.7 Subject term (key word) listing.

Ammunition
Tank
Cannon
Cartridge
Case, Cartridge
Spiral-Wrapped

Custodian:
ARMY-AR

Preparing activity:
ARMY-AR

Project No: 1315-AC08

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70950 (AR)	CASE, CARTRIDGE, 105MM, M148A1B1E1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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